

Subject : MD - I

Name : Wagh Trupti Dnyaneshwar

Class : T.Y. Btech

Batch: T3

Roll No.: 51

Expt. No.:

Date : / / 202

Division: -

Title : Que. Paper - 2023 Winter

Remark

Signature

[Q.1]

A) Write answers of following mcq.

1) Components made of brittle materials fails because of
→ Sudden Fracture.

2) The items produced by craft evolution process.
→ Rowing boat, Musical instruments, Bullock cart.

3) Ergonomics means
→ Natural Laws of Work.

4) A sunk key fits in the keyway of the
→ Both hub & sleeve

5) Woodruff key permits movement between
Shaft & the hub.
→ Radial.

6) A tension member of 40mm diameter is to be replaced by a square of same material, side of square will be
→

B) ~~1~~ Distinguish between design synthesis & design analysis.

→ a) Design Synthesis:

- i) It is a phase where designer generate & develop creative ideas & concepts to address specific design problem.
- ii) It comes with innovative solutions to meet a set of requirement.
- iii) During design synthesis ideate, sketch & create prototypes to explore various design possibilities.
- iv) It's primary outcome is a collection of design concepts or proposals that represent potential solutions to the problem at hand.

b) Design Analysis:

- i) It is a phase where designers evaluate & assess the propose design concepts to ensure they meet the necessary criteria, standards & performance requirements.
- ii) It involves a systematic examination of the feasibility & viability of the design ideas.
- iii) It analyze how well the design concepts align with the project goals & constraints.
- iv) The primary outcome of design analysis is a set of findings & recommendations.

② What is standardization? Explain its importance in design.

→ Standardization refers to the process of developing and implementing consistent & uniform practices, specifications or guidelines for various processes, products or services across different industries & organizations. Standardization can apply to various fields, including manufacturing, technology, health-care & more.

Importance:

- 1) Quality Assurance: Standardization helps to ensure the quality & reliability of products & services.
- 2) Interoperability: In technology & various industries standardization is crucial for enabling interoperability.
- 3) Safety: Standardization also includes safety requirements & regulations.
- 4) Cost Reduction: Standardization can lead to cost savings by streamlining processes, reducing waste & minimizing the need for custom solutions.
- 5) Consumer Confidence: Consumers benefit from standardization by knowing that products & services meet certain quality & safety criteria.

c) Write short note on:

1) Aesthetic consideration:

Aesthetic consideration is defined as a set of principles of appreciation of beauty. It deals with the appearance of the product. Appearance is an outward expression of quality of the product & is the first communication of the product with user. These consideration are highly influenced by individual preference, cultural norms & historical contexts.

2) Factor of Safety & its selection:

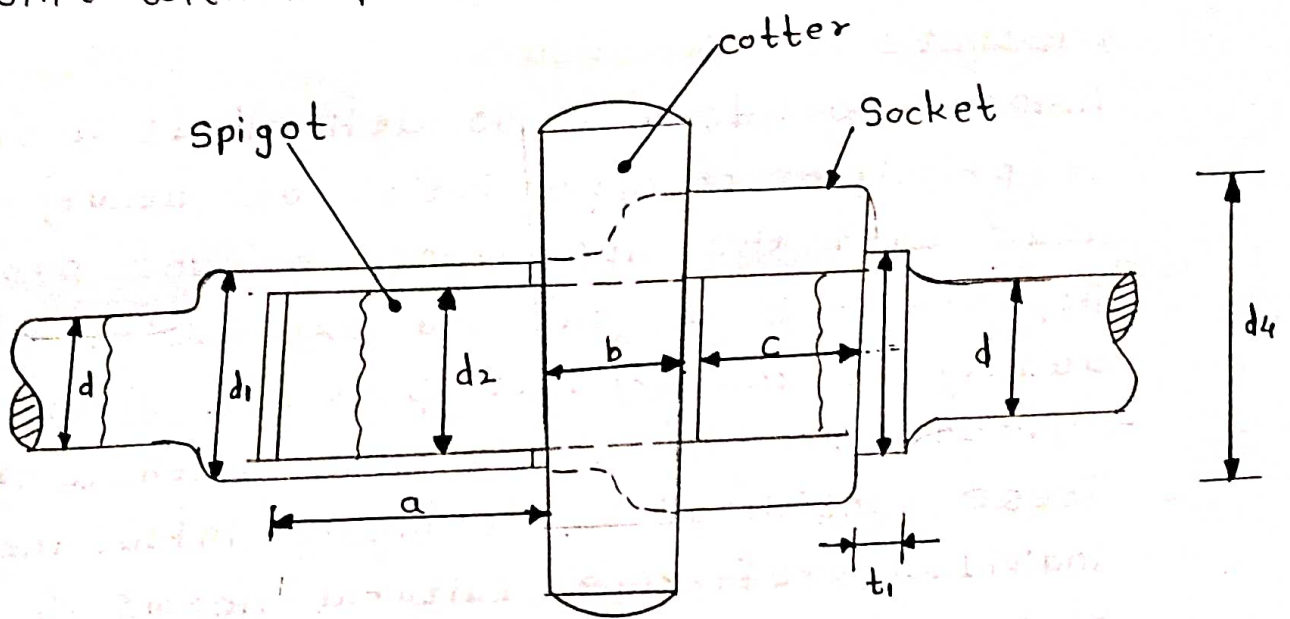
It is a critical engineering concept used to ensure the safety & reliability structures, components & systems. It represent the ratio of maximum allowable stress or load material can withstand actual applied stress or load.

The selection of Factor of Safety includes:

- ① Purpose of Factor of Safety
- ② Determining factor of safety
- ③ Factors influencing FOS selection.
- ④ Codes & standards
- ⑤ Optimization.

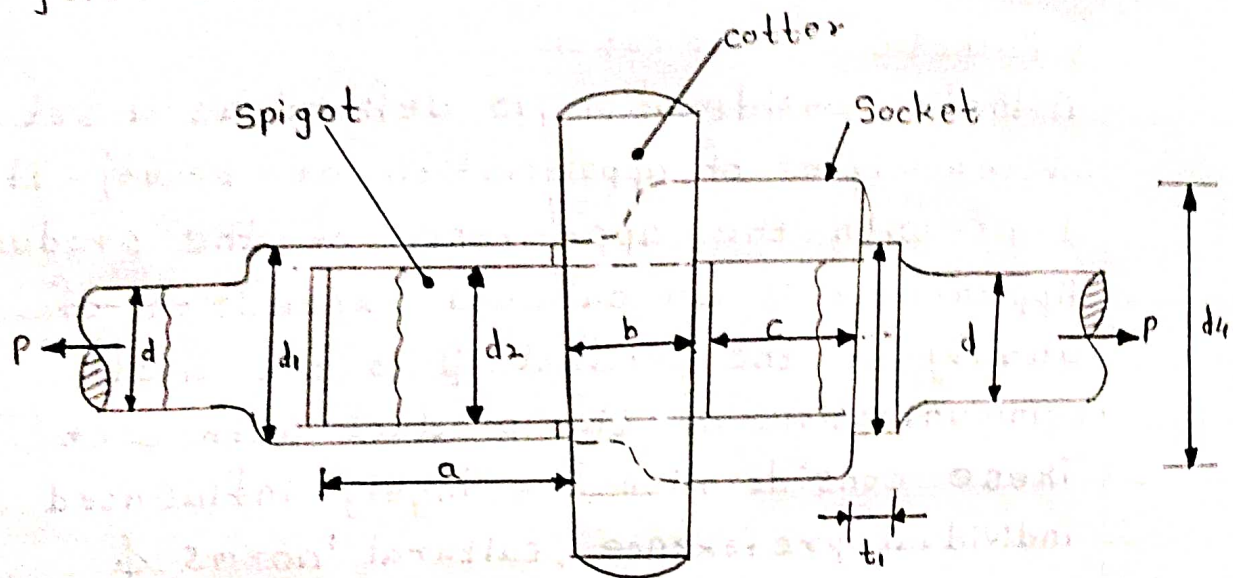
[Q.2]

A) Explain construction & working principle of cotter joint with help of neat diagram.



[Q.2]

A) Explain construction & working principle of cotter joint with help of neat diagram.



Construction :

- 1) Socket : The socket is usually a hollow cylinder that accommodates the spigot.
- 2) Cotter : The rod with the spigot end is inserted in socket cavity.
- 3) Spigot : It is the male portion of the joint & a rectangular slot allows the cotter to be inserted through it.

Working Principles :

- 1) Assembly : To assemble the cotter joint, the spigot is inserted in cotter joint.
- 2) Axial Load Transfer : Once cotter joint is assembled, it can transmit axial loads from one component to another.
- 3) Limited Angular Movement : Cotter joint allows for a limited amount of angular movements between the connected components.

B) The stress induced at critical point in a m/c component made of 45C8 [$S_{yt} = 380 \text{ N/mm}^2$] are as follows.

$$\sigma_x = 100 \text{ N/mm}^2$$

$$\sigma_y = 40 \text{ N/mm}^2$$

$$\tau = 80 \text{ N/mm}^2$$

Calculate FOS by

① Max^m. normal stress theory

② Max^m. shear stress theory

③ The distortion energy theory

→ Given: $\sigma_x = 100 \text{ N/mm}^2$

$$\sigma_y = 40 \text{ N/mm}^2$$

$$\tau = 80 \text{ N/mm}^2$$

$$S_{yt} = 380 \text{ N/mm}^2$$

① Max^m. normal stress theory:

$$\frac{S_{yt}}{\text{FOS}} = \frac{\sigma_x + \sigma_y}{2} + \frac{1}{2} \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau^2}$$

$$\frac{380}{\text{FOS}} = \frac{100 + 40}{2} + \frac{1}{2} \sqrt{(100 - 40)^2 + 4 \times 80^2}$$

$$\boxed{\text{FOS} = 2.44}$$

② Max^m. shear stress theory:

$$\frac{0.5 \times S_{yt}}{\text{FOS}} = \frac{1}{2} \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau^2}$$

$$\frac{0.5 \times 380}{\text{FOS}} = \frac{1}{2} \sqrt{(100 - 40)^2 + 4 \times 80^2}$$

$$\boxed{\text{FOS} = 2.22}$$

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Calculate FOS by

(i) Max^m. normal stress theory

(ii) Max^m. shear stress theory

(iii) The distortion energy theory

→ Given: $\sigma_x = 100 \text{ N/mm}^2$

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$$S_{yt} = 380 \text{ N/mm}^2$$

(i) Max^m. normal stress theory:

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(ii) Max^m. shear stress theory:

$$\frac{0.5 \times S_{yt}}{\text{FOS}} = \frac{1}{2} \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau^2}$$

$$\frac{0.5 \times 380}{\text{FOS}} = \frac{1}{2} \sqrt{(100 - 40)^2 + 4 \times 80^2}$$

$$\boxed{\text{FOS} = 2.22}$$

③ The distortion energy Theory:

— Max^m. principle strain energy theory:

$$\sigma_{t1} = \frac{\sigma_x + \sigma_y}{2} + \frac{1}{2} \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau^2}$$

$$\sigma_{t1} = \frac{100 + 40}{2} + \frac{1}{2} \sqrt{(100 - 40)^2 + 4 \times 80^2}$$

$$\sigma_{t1} = 155.44 \text{ N/mm}^2$$

$$\sigma_{t2} = \frac{\sigma_x + \sigma_y}{2} - \frac{1}{2} \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau^2}$$

$$\sigma_{t2} = \frac{100 + 40}{2} - \frac{1}{2} \sqrt{(100 - 40)^2 + 4 \times 80^2}$$

$$\sigma_{t2} = 70 \text{ N/mm}^2$$

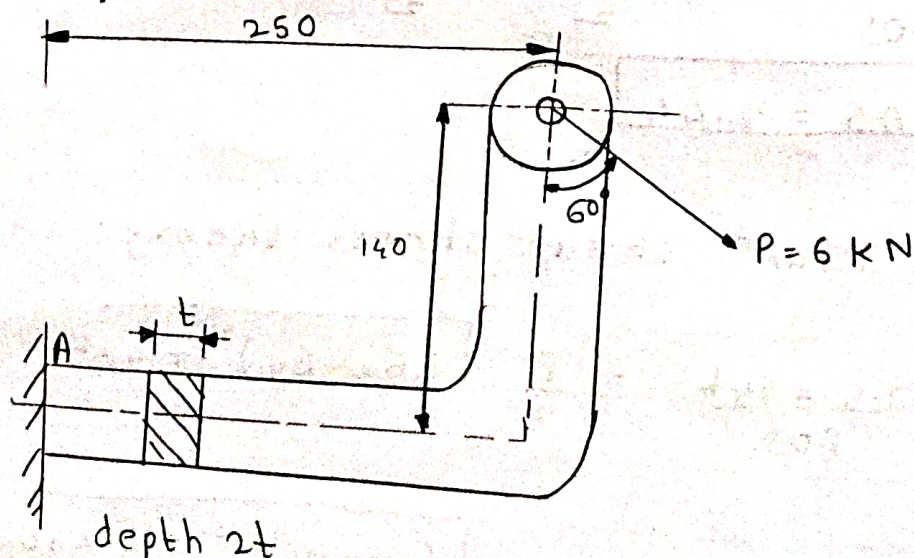
Now, by using max^m. distortion energy theory:

$$\sigma_{t1}^2 + \sigma_{t2}^2 - 2\sigma_{t1} \times \sigma_{t2} = \left(\frac{S_{yt}}{FOS} \right)^2$$

$$155.44^2 + 70^2 - 2 \times 155.44 \times 70 = \left(\frac{380}{FOS} \right)^2$$

$$FOS = 4.44$$

c)



→ Given: $P = 6 \text{ kN} = 6 \times 10^3 \text{ N}$

$\text{FOS} = 3$

Assume max^m. normal stress theory of failure.

$P_h = (6 \times 10^3) \sin 60 = 5196.15 \text{ N}$

$P_v = (6 \times 10^3) \cos 60 = 3000 \text{ N}$

$M = (P_h \times 140) + (P_v \times 250)$

$M = (5196.15 \times 140) + (3000 \times 250)$

$M = 1.47 \times 10^6 \text{ N.mm}$

$y = \frac{zt}{z} = t$

$I = \frac{bd^3}{12} = \frac{t(2t)^3}{12} = \frac{8t^4}{12} = 0.66t^4$

$\sigma_b = \frac{My}{I}$

$\sigma_b = \frac{(1.47 \times 10^6) \times t}{0.66t^4}$

$\sigma_b = \frac{1.47 \times 10^6}{0.66t^3}$

$\sigma_b = \frac{2.22 \times 10^6}{t^3} \quad \text{--- (1)}$

$\sigma_t = \frac{P_h}{2t \times t} = \frac{5196.15}{2t^2} = \frac{2598.07}{t^2}$

$$G = G_b + G_t$$

$$\frac{FG}{FOS} = \frac{2.22 \times 10^6}{t^3} + \frac{2598.07}{t^2}$$

$$\frac{250}{3} = \left[\frac{2.22 \times 10^6}{t^3} \right] + \left[\frac{2598.07}{t^2} \right]$$

$$t = 30.21 \text{ mm}$$

[Q.3]

A) ① What is stress concentration? Explain various methods of reducing stress concentration.

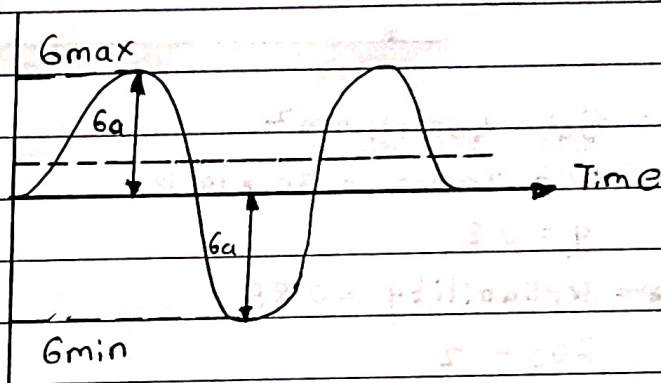
→ Stress concentration also known as stress risers. It is a phenomenon in mechanics where localized area in a component experience significantly higher stress levels compared to surrounding material.

Various Methods:

- i) Fillet or Radius: Fillets distribute the stress more smoothly, ~~or~~ reducing stress concentration.
- ii) Smooth Transitions: Avoid abrupt changes in geometry such as sharp corners or sudden shifts in cross-sectional area.
- iii) Change in material properties: Selecting material for better fatigue resistance or mechanical properties can reduce stress concentration.
- iv) Notch Sensivity Reduction: It may be necessary to modify the design to minimize notches.

2) What is fluctuating stress? Draw stress-time curve for fluctuating stress.

→ Fluctuating stress are also known as cyclic stress. It is a type of stress that varies in magnitude over time due to cyclic or repetitive loading on a material. This cyclic loading occur due to various mechanical factors, including vibrations, thermal cycling, repeated application of forces.



3) Explain the terms of fatigue failure & fatigue life?

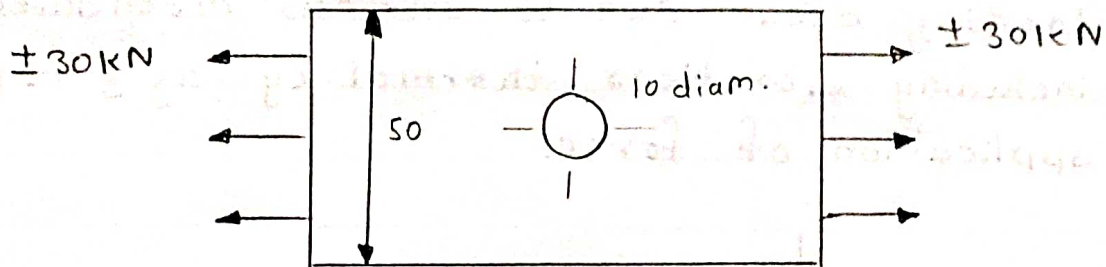
→ (a) Fatigue failure :

Fatigue failure is the process by which a material fails under cyclic load at stress levels significantly below their ultimate or yield strength. Fatigue failure can occur in a variety of components, including machinery, bridges, aircraft & more. These failures are typically initiated by microscopic defects, stress concentration.

⑥ Fatigue Life:

Fatigue life, also known as the fatigue endurance or fatigue limit, is the no. of stress or strain cycles that a material or component can endure before it experiences fatigue failure. The fatigue life is determined through fatigue testing.

B)



→ Given: $S_{ut} = 400 \text{ N/mm}^2$

$$P = 30 \text{ kN} = 30 \times 10^3 \text{ N}$$

$$q = 0.8$$

$$k_a = \text{Reliability} = 0.85$$

$$\text{FOS} = 2$$

$$k_a = 0.67, k_b = 0.85, k_c = 0.897, k_t = 2.51$$

$$① S_e' = 0.5 S_{ut} = 0.5 \times 400 = 200 \text{ MPa}$$

$$② k_d = \frac{1}{k_f}$$

$$\text{but, } k_f = q(k_t - 1) + 1$$

$$k_f = 0.8(2.51 - 1) + 1$$

$$k_f = 2.208$$

$$k_d = 0.452$$

$$③ S_e = k_a \cdot k_b \cdot k_c \cdot k_d \cdot k_e \cdot k_g \cdot S_e'$$

$$S_e = 0.67 \times 0.85 \times 0.897 \times 0.452 \times 1 \times 1 \times 200$$

$$S_e = 46.18 \text{ MPa}$$

$$\textcircled{4} \quad G_a = \frac{S_e}{FOS} = \frac{46.18}{2} = 23.09 \text{ MPa.}$$

$$\textcircled{5} \quad G_a = \frac{P}{(w-d) \times t}$$

$$23.09 = \frac{30 \times 10^3}{(50-10) \times t}$$

$$t = 32.48 \text{ mm}$$

$$④ \quad G_a = \frac{S_e}{FOS} = \frac{46.18}{2} = 23.09 \text{ MPa.}$$

$$⑤ \quad G_a = \frac{P}{(w-d) \times t}$$

$$23.09 = \frac{30 \times 10^3}{(50-10) \times t}$$

$$t = 32.48 \text{ mm}$$

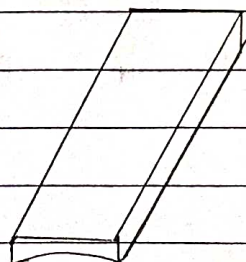
[Q.4]

A) What is the function of keys? Explain the different types of keys along with their neat sketches.

→ keys used as a connecting element between a rotating element, such as shaft & a mating part like a gear or pulley. The key prevents relative motion between the two parts, ensuring that they rotate together.

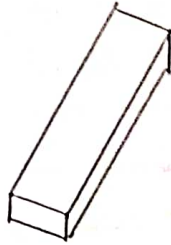
① Saddle key:

It is also known as a sunk saddle key, its similar to a sunk key but has a slight curvature on the top surface. This design helps to distributing more stresses ^{evenly} ~~evenly~~.



② Sunk key:

This is a rectangular or square key that fits into the keyways cut into both the shaft & the hub of the rotating element. It's one of the simplest types of keys.



③ Woodruff key:

This is a half-moon shaped key that fits into a curved keyseat in the shaft.



B) Given: $S_{yt} = 380 \text{ N/mm}^2$

diam. of pulley = 1000 mm

$$F_1 = 3 \text{ kN}$$

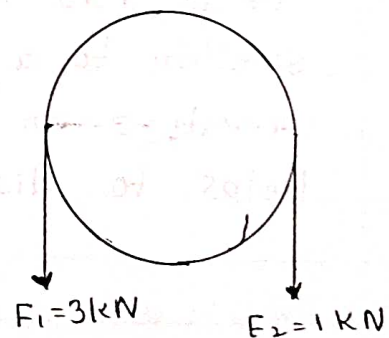
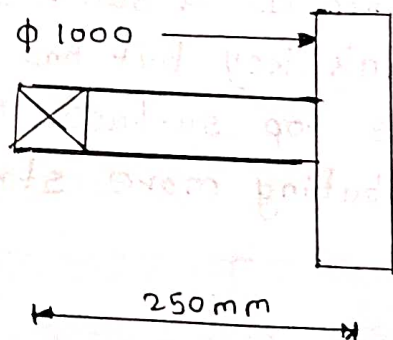
$$F_2 = 1 \text{ kN}$$

$$W = 500 \text{ N}$$

$$\text{FOS} = 3$$

$$d = ?$$

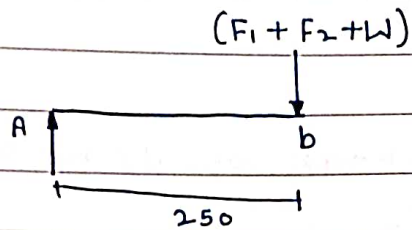
Solution:



$$T = (F_1 - F_2) \times r$$

$$T = ((3 \times 10^3) - (1 \times 10^3)) \times 500$$

$$T = 1 \times 10^6 \text{ N.m}$$



$$M_b = (F_1 + F_2 + W) \times 250$$

$$M_b = (3000 + 1000 + 500) \times 250$$

$$M_b = 1.125 \times 10^6 \text{ N.m}$$

$$\tau = \frac{S_{yt}}{2FOS} = \frac{380}{2 \times 3} = 63.33 \text{ N/mm}^2$$

$$\tau = \frac{16}{\pi d^3} \times T_e \quad \dots \quad T_e = \sqrt{T^2 + M_b^2}$$

$$63.33 = \frac{16}{\pi d^3} \times \sqrt{(1 \times 10^6)^2 + (1.125 \times 10^6)^2}$$

$$d = 49.41 \text{ mm}$$

c) Given: $P = 15 \times 10^3 \text{ W}$ 15 kW

$$N = 100 \text{ rpm}$$

$$n = 6, d = 200$$

$$\tau = 35 \text{ N/mm}^2$$

$$\sigma_b = 152 \text{ N/mm}^2$$

$$(l)_b = 35 \text{ mm}$$

① Pin diameter by shear consideration:

$$M_t = \frac{[60 \times 10^6] \times [P]}{2\pi N}$$

$$M_t = \frac{[60 \times 10^6] \times [15 \times 10^3]}{2 \times \pi \times 100}$$

$$(M_t) = 1.43 \times 10^6 \text{ N.mm}$$

$$\tau = \frac{8M_t}{\pi d_1^3 \times n} \Rightarrow \tau = \frac{8(M_t)}{\pi d_1^3 \times P \times n}$$

$$35 = \frac{8 \times (1.43 \times 10^6)}{\pi \times d_1^3 \times 200 \times 6}$$

$$\boxed{d_1 = 9.31 \text{ mm}}$$

② Pin diam. by bending consideration

$$(M_t)_i = P \times \frac{D}{2} \times n$$

$$P = \frac{2 \times (M_t)_i}{P \times n} = \frac{2 \times (1.43 \times 10^6)}{200 \times 6} = 2383.33 \text{ N}$$

$$M_b = P \left[5 + \frac{35}{2} \right] \rightarrow (l)_b$$

$$M_b = 2383.33 \times 22.5$$

$$M_b = 53624.92 \text{ N.mm}$$

$$6b = \frac{32Mb}{\pi d_2^3}$$

$$152 = \frac{32 \times 53624.92}{\pi d_2^3}$$

$$d_2 = 15.31 \text{ mm}$$

[Q.5]

A) (i) What is the cause of residual stresses in welded joint? How they are relieved?

→ (i) - (i) Non-uniform Heating & Cooling:

- Rapid heating & cooling.
- Thermal contraction & expansion

(ii) Phase Transformation:

- Metallurgical Changes.

(iii) Constrained Thermal expansion:

- Restraint from surrounding material.

(iv) Volume changes:

- Shrinkage

(v) Joint Geometry:

- Joint Design

(vi) Welding Process Parameters:

- Welding speed
- Heat input.

- (vii) Material Properties:
- Thermal Properties
 - Coeff. of Thermal expansion.

(viii) How they relieved?

(i) Temp. & duration:

This is critical factor. The treatment is typically carried out at temp. below critical transformation.

(ii) Critical Cooling:

Control cooling phase helps to prevent the reinduction of new stresses.

(iii) Stress distribution:

The heat treatment allows the material to reach a state of equilibrium, redistributing stresses & minimizing potential for distortion & cracking.

2) Explain the surge phenomena in spring & methods to avoid same.

→ The surge phenomena in springs refers to an undesired effect where the spring exhibits sudden & excessive displacement or deformation beyond the desired range. This surge can lead to various issues, including potential damage to spring. The surge phenomenon is often associated with dynamic loading conditions, such as rapid changes in force or sudden impacts.

Methods to Avoid surge Phenomenon:

1. Damping
2. Frequency Analysis
3. Load Limiting
4. Gradual Loading
5. Stiffer spring
6. Prevent Resonance
7. Material Selection
8. Regular Maintenance.

B) Given: $P = 3.5 \times 10^3 \text{ N}$

$$\delta = 5 \text{ mm}$$

$$C = 5$$

$$G = 81370 \text{ N/mm}^2$$

$$S_{ut} = 1050 \text{ N/mm}^2$$

$$\tau = 50\% S_{ut}$$

→ ① Wire diameter:

$$k = \frac{4C-1}{4C-4} + \frac{0.615}{C}$$

$$k = \frac{4(5)-1}{4(5)-4} + \frac{0.615}{5}$$

$$k = 1.3105$$

$$\tau = 0.5 S_{ut}$$

$$\tau = 0.5 \times 1050$$

$$\tau = 525 \text{ N/mm}^2$$

$$\tau = k \left[\frac{8PC}{\pi d^2} \right]$$

$$525 = 1.3105 \left[\frac{8 \times (3.5 \times 10^3) \times 5}{\pi d^2} \right]$$

$$d = 9.21 \text{ mm}$$

② Mean coil diameter:

$$D = Cd$$

$$D = 5 \times 9.21$$

$$D = 46.05 \text{ mm}$$

③ No. of active coils

$$\delta = \frac{8PD^3N}{Gd^4}$$

$$5 = \frac{8 \times (3.5 \times 10^3) \times 46.05^3 \times N}{81370 \times 9.21^4}$$

$$N = 1.07 \approx 1$$

④ Total no. of coils

$$N_t = N + 2 = 1 + 2 = 3$$

⑤ Solid length of spring

$$l = N_t \cdot d = 3 \times 9.21 = 27.63 \text{ mm}$$

⑥ Free length of spring

$$\delta = \frac{8PD^3N}{Gd^4}$$

$$\delta = \frac{8 \times (3.5 \times 10^3) \times 46^3 \times 1}{81370 \times 9.21^4}$$

$$\delta = 4.64 \text{ mm}$$